Kolla Sai Yashaswini

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2015	6.23
Sri Chaitanya Junior Kalasala, Bagh Amberpet, Hyderabad	Class XII (Telangana State Board of Intermediate Education)	2023	97.20%
Sri Chaitanya Techno School, L.B. Nagar, Ranga Reddy	Class X (Board of Secondary Education, Telangana)	2021	10.00%

PROJECTS

Full-Stack Web Application

- Developed a Java backend with Spring Boot and PostgreSQL, and built a React frontend that connects to it through web services.
- Containerized the application with Docker and deployed it to Amazon Web Services at a public web address.
- Automated testing and deployment with a GitHub Actions pipeline on every code update.

Agentic Finance Research Assistant

- Built a retrieval-augmented question-answering web app in Python and FastAPI that returns cited investment memos from uploaded company reports.
- Designed a multi-agent pipeline (retrieval, analysis, report) in which code, not the language model, generates the source list, so every citation traces to a real document.
- Developed an automated groundedness check that flags any figure or citation in a memo that is missing from the retrieved evidence.
- Evaluated keyword (TF-IDF) retrieval against embedding-based retrieval and reported the results honestly.

Custom Programming Language Interpreter (C++)

- Designed and implemented a tree-walking interpreter for a custom programming language, including a lexer, recursive-descent parser, and AST-based evaluator
- Implemented core language features: variable scoping, control flow (if/while), and function calls with closures
- Built a REPL for interactive execution and a test suite covering lexer, parser, and evaluator edge cases
- Documented language grammar (EBNF) and design decisions; integrated CI via GitHub Actions for automated testing

SDN Load Balancer Simulation

- Built a software-defined networking simulation of a load balancer using Mininet, supporting 6 load balancing algorithms across 8 backend servers.
- Implemented a multi-threaded HTTP key-value store and client simulator supporting up to 50 concurrent clients, with GET, PUT, and DELETE operations.
- Benchmarked average response time across concurrency levels (1-50 clients) and logged results for performance comparison across load balancing strategies.

SKILLS

- **Programming:** C++, Python, Java, SQL
- **Core CS:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks
- **Systems & Tools:** Linux, Multithreading & Concurrency, Git, Docker

- **AI/ML:** Machine Learning Basics, Retrieval-Augmented Generation, Text Embeddings, LLM APIs

RELEVANT COURSES

- **Data & Analytics:** Probability & Random Variables, Statistics, Introduction to AI/ML, DBMS-I.
- **Computer Science:** Data Structures & Algorithms, Theory of Computation, Computer Networks, OS-I; Ongoing -- Linear optimisation.
- **Mathematics:** Discrete Mathematics, Differential Equations, Elementary Linear Algebra, Transform Techniques, Number Theory.

SCHOLASTIC ACHIEVEMENTS

- Qualified NTSE (National Talent Search Examination) Stage 1, conducted by NCERT -- one of India's most prestigious scholarship programs.
- Qualified JEE Main & JEE Advanced, conducted by NTA -- India's most competitive engineering entrance examinations.
- Secured 3rd place in Film Making, Milan Inter-Hostel Championship, IIT Hyderabad.

POSITIONS OF RESPONSIBILITY

Core Member, Behind The Lens (Filmmaking Club), IIT Hyderabad

- Coordinated marketing events and campus promotions, and contributed to short-film productions.

Design Coordinator, Finance and Consulting Club, IIT Hyderabad

- Designed event posters, social media creatives and promotional materials to increase club outreach.
- Directed visual branding with the core team, keeping every design consistent and aligned with club objectives.

Souvenir Head, Alumni Cell, IIT Hyderabad

- Managed alumni communication across email, social media and college departments.
- Co-led alumni engagement covering reunions, guest lectures and mentorship sessions, coordinating cross-functional teams to execute logistics

EXTRACURRICULAR

- Active NSS volunteer; digital arts and portrait sketching